Networking Revision

IE3122 – Network Security | Week 1

A quick refresher before we build on it

What We're Refreshing

- Layered architecture (OSI & TCP/IP)
- Switching & routing fundamentals
- Ethernet & IP addressing
- DHCP & ARP
- TCP & UDP
- HTTP

Why revisit this?

Every attack and defense mechanism in this module sits on top of these fundamentals — you can't secure a layer you don't understand.

This is a refresher, not first-time teaching — we move quickly and focus on what matters for security.

How This Fits the Module

- Underpins LO1 (describing vulnerabilities) and LO3 (analysing network/communication requirements).
- Weeks 3–5 build directly on this: Layer 2, 3, and 4 attacks all exploit gaps in the protocols we're reviewing today.
- If any of this feels shaky, flag it early — it only gets more security-focused from here.

Layered Architecture

Why we split networking into layers

CONCEPT

Why Layered Architecture?

- Breaks a complex problem (moving data across a network) into independent, manageable pieces.
- Each layer has one job and only talks to the layers directly above and below it.
- Vendors can innovate within a layer (e.g. a new Ethernet standard) without breaking the layers above.
- Security matters: attackers target specific layers, so defenders need layer-by-layer thinking.

CONCEPT

The OSI 7-Layer Model

7	Application	*HTTP, DNS, SMTP*
6	Presentation	*Encryption, encoding*
5	Session	*Session mgmt*
4	Transport	*TCP, UDP*
3	Network	*IP, routing*
2	Data Link	*Ethernet, MAC, ARP*
1	Physical	*Cabling, signals*

CONCEPT

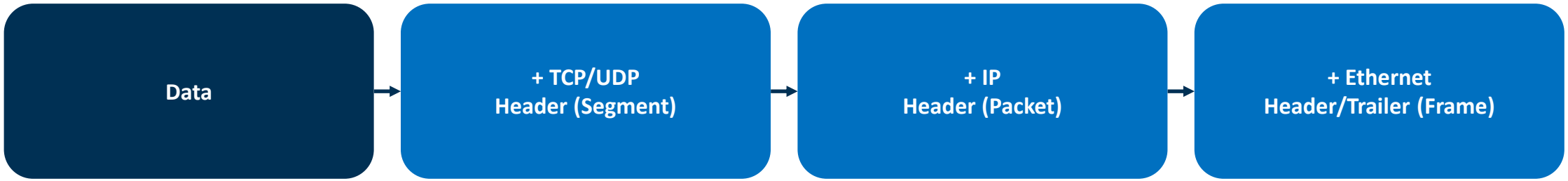
OSI vs TCP/IP Model

OSI (7 layers)	TCP/IP (4 layers)
Application / Presentation / Session	Application
Transport	Transport
Network	Internet
Data Link / Physical	Network Access

TCP/IP is what's actually implemented on the Internet; OSI is the conceptual teaching model we use to reason about it.

CONCEPT

Encapsulation & Decapsulation



- Encapsulation: as data moves down the stack (sender), each layer wraps it with its own header (and sometimes trailer).
- Decapsulation: as it moves up the stack (receiver), each layer strips off and reads its own header.
- Security relevance: many attacks (spoofing, injection) work by tampering with a specific layer's header.

Switching Revision

Layer 2 forwarding fundamentals

What Switches Do

- Operate at Layer 2 — forward frames based on destination MAC address.
- Build and maintain a MAC address table by learning source MACs on incoming frames.
- Unknown destination → flood out all ports except the incoming one; known destination → forward out one port only.
- Each switch port is its own collision domain — this is why switches replaced hubs.

Domains & VLANs

- Collision domain: a segment where frames can collide — each switch port is its own collision domain.
- Broadcast domain: the set of devices that receive a broadcast — by default, all ports on a switch share one broadcast domain.
- VLANs (Virtual LANs) split a switch into multiple broadcast domains for segmentation — a core building block for later security topics like zone-based defenses.

Common Switching Issues

- Switching loops: redundant links can cause broadcast storms — prevented by Spanning Tree Protocol (STP).
- MAC table exhaustion / CAM overflow: an attacker floods fake MACs to force a switch into flooding all frames (foreshadows Week 4–5 Layer 2 attacks).
- We'll dig into these as attacks — and their defenses — later in the module.

Routing Revision

Layer 3 forwarding fundamentals

What Routers Do

- Operate at Layer 3 — forward packets based on destination IP address.
- Maintain a routing table mapping networks to the best next-hop interface.
- Connect different networks/broadcast domains together — a router is a natural boundary for security policy.
- Every host needs a default gateway — the router it sends traffic to when the destination isn't on its local network.

Static vs Dynamic Routing

	Static Routing	Dynamic Routing
Configuration	Manually entered by admin	Learned automatically via a routing protocol
Best for	Small, stable networks	Larger networks with changing topology
Security angle	Predictable, smaller attack surface	Vulnerable to route injection/spoofing if unauthenticated

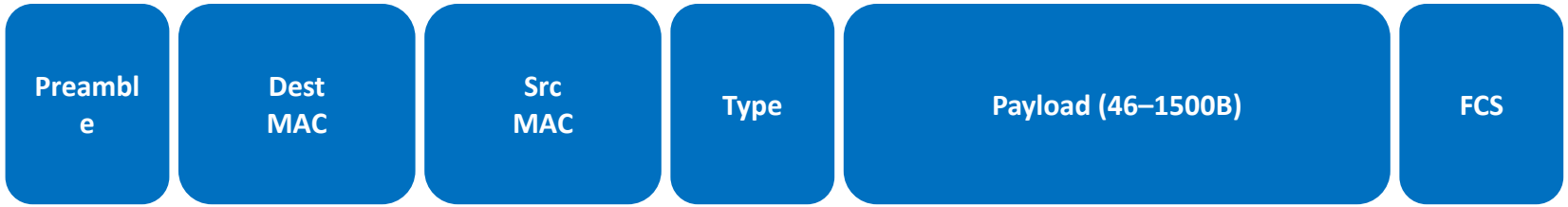
Routing Protocols — Quick Overview

Protocol	Type	Notes
RIP	Distance-vector	Simple, hop-count based, limited scalability
OSPF	Link-state	Fast convergence, widely used internally
EIGRP	Advanced distance-vector	Cisco-developed, fast convergence
BGP	Path-vector	Routes the Internet itself, between autonomous systems

Ethernet

The dominant Layer 2 / Layer 1 technology

The Ethernet Frame



- Destination/Source MAC: 6-byte hardware addresses identifying sender and receiver on the local segment.
- Type: identifies the encapsulated protocol (e.g. IPv4, ARP).
- FCS (Frame Check Sequence): CRC checksum for error detection — not tamper protection.

MAC Addressing

- 48-bit address, usually written as six hex pairs (e.g. 00:1A:2B:3C:4D:5E).
- First 3 bytes (OUI) identify the manufacturer; last 3 bytes are the device-specific identifier.
- Burned into hardware, but can be spoofed in software — a common Layer 2 attack technique we'll cover later.

Ethernet Standards & Speeds

Standard	Speed	Common Use
10BASE-T	10 Mbps	Legacy
100BASE-TX (Fast Ethernet)	100 Mbps	Older LANs
1000BASE-T (Gigabit)	1 Gbps	Standard LAN today
10GBASE-T and above	10 Gbps+	Data centers, backbones

IP

Logical addressing and routing at Layer 3

IPv4 Addressing

- 32-bit address, written as 4 decimal octets (e.g. 192.168.1.10).
- Split into network and host portions, defined by a subnet mask (e.g. /24 = 255.255.255.0).
- Address classes (A, B, C) are largely historical — modern networks use CIDR (Classless Inter-Domain Routing).
- Private ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) aren't routed on the public Internet — relevant to NAT and perimeter security.

Subnetting Basics

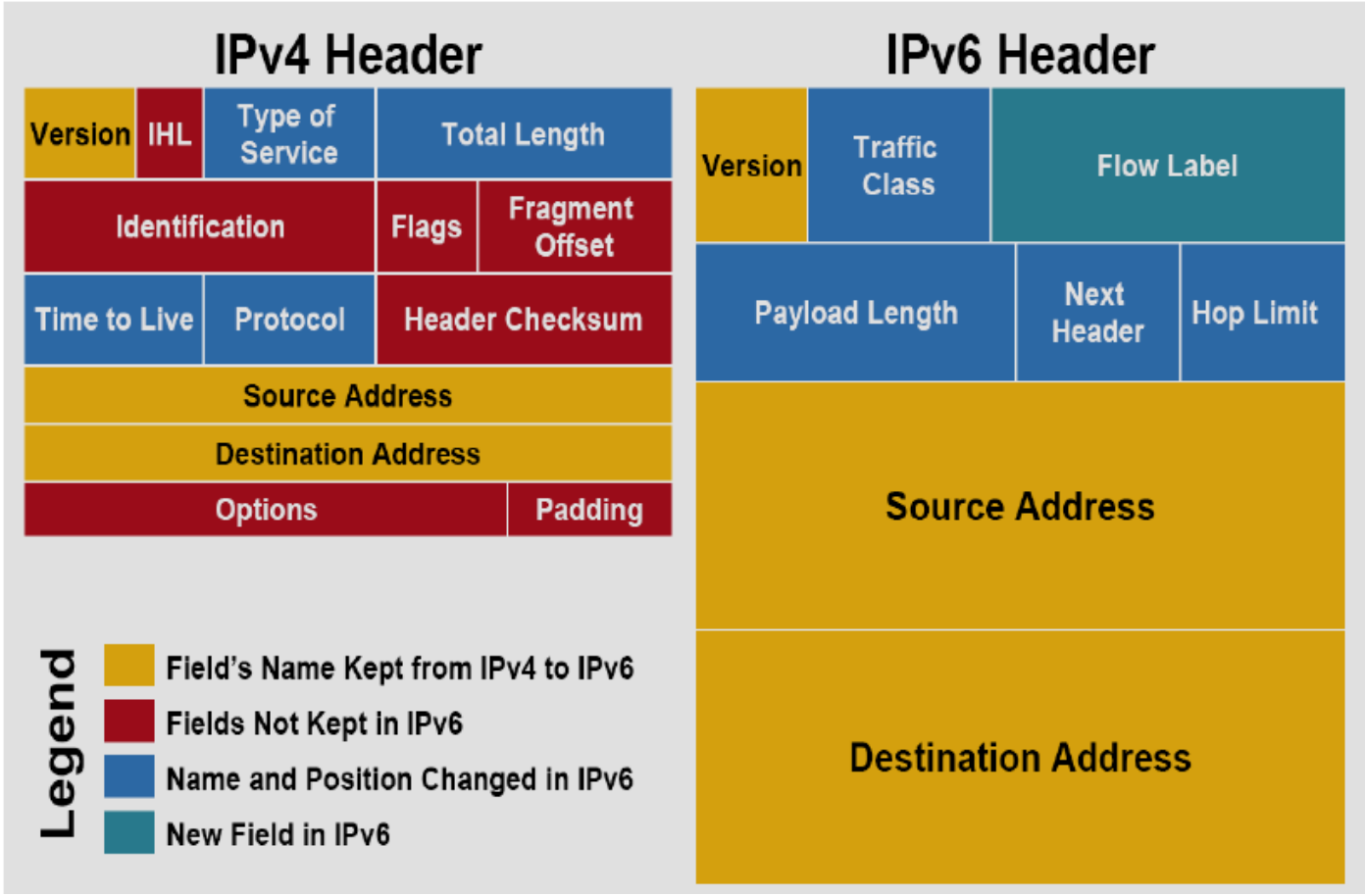
CIDR	Subnet Mask	Usable Hosts
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/30	255.255.255.252	2 (point-to-point links)

Subnetting isn't just addressing — smaller subnets mean smaller broadcast domains and tighter security boundaries.

IPv4 vs IPv6

	IPv4	IPv6
Address length	32-bit	128-bit
Address space	~4.3 billion	3.4×10^{38}
Built-in security	None (IPSec optional)	IPSec support built into the standard
Notation	192.168.1.10	2001:0db8::1

IPv4 vs IPv6 Header



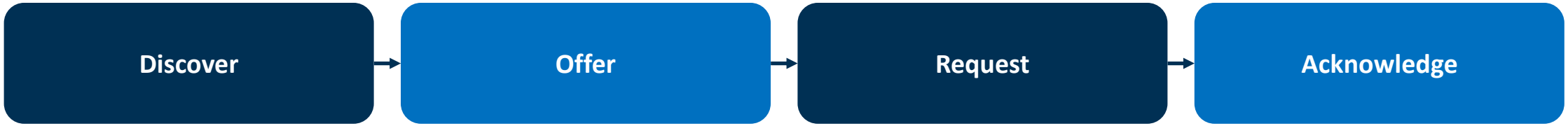
IP Header — Key Fields

- TTL (Time to Live): limits how many hops a packet can traverse — prevents infinite routing loops.
- Protocol: identifies the transport-layer protocol carried (TCP=6, UDP=17, ICMP=1).
- Source/Destination IP: the addresses routers use to forward the packet.
- Fragmentation flags/offset: support splitting packets too large for a given link — also a historical attack vector (fragmentation attacks).

DHCP

Automatic IP address assignment

DHCP: DORA Process



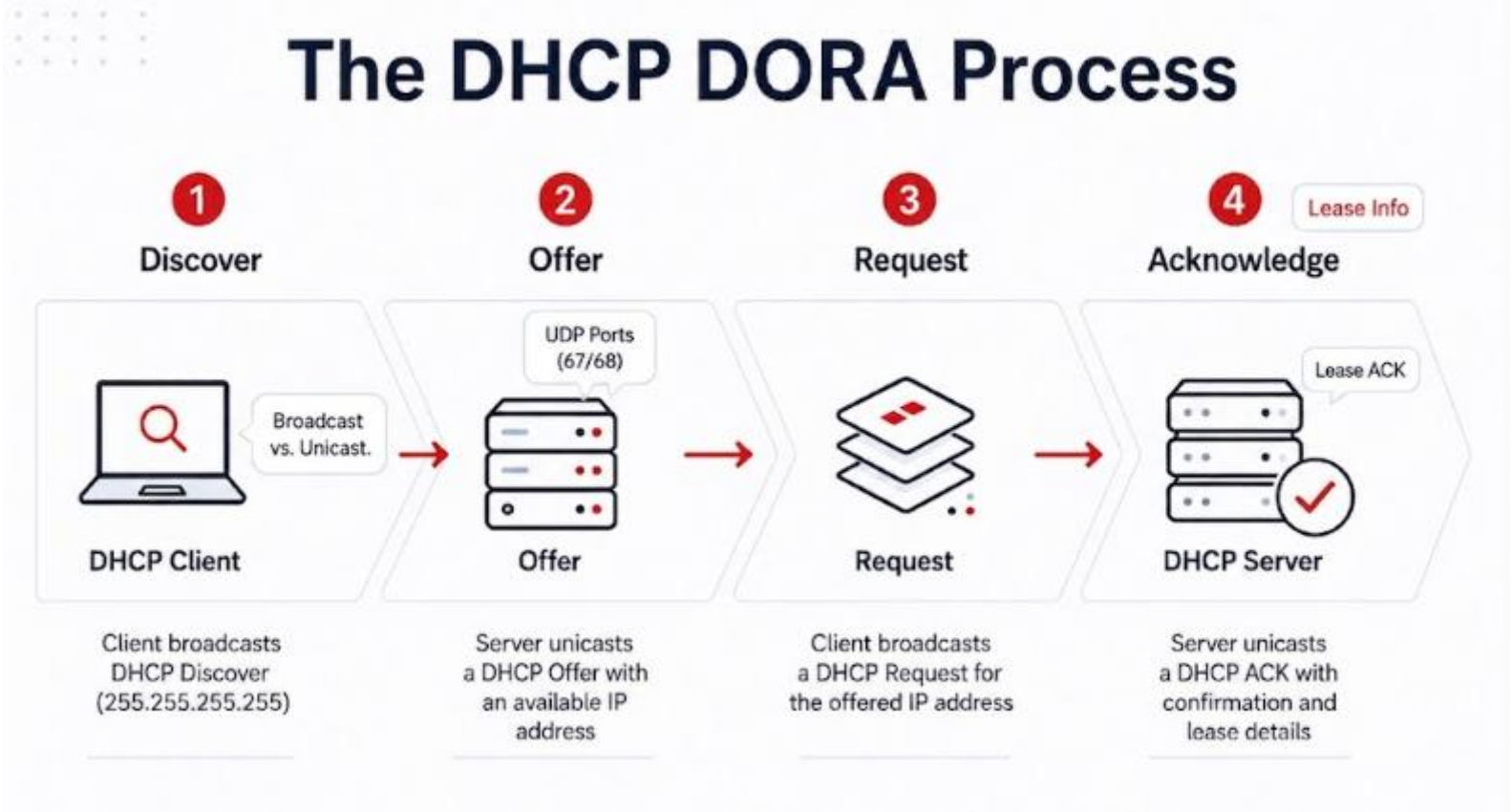
Client broadcasts for a DHCP server

Server offers an available IP

Client requests the offered IP

Server confirms the lease

DHCP: DORA Process



DHCP Details

- Uses UDP ports 67 (server) and 68 (client).
- Leases are time-limited — clients must renew before expiry or risk losing their address.
- DHCP options can also hand out gateway, DNS servers, and other config — not just an IP address.

DHCP: A Security Preview

- DHCP has no built-in authentication — any device can respond to a Discover.
- Rogue DHCP server attack: an attacker's server races the real one to hand out a malicious gateway/DNS, redirecting victim traffic.
- Mitigation preview: DHCP snooping on switches (we'll cover this properly under Layer 2 attacks & network defense).

ARP

Mapping IP addresses to MAC addresses

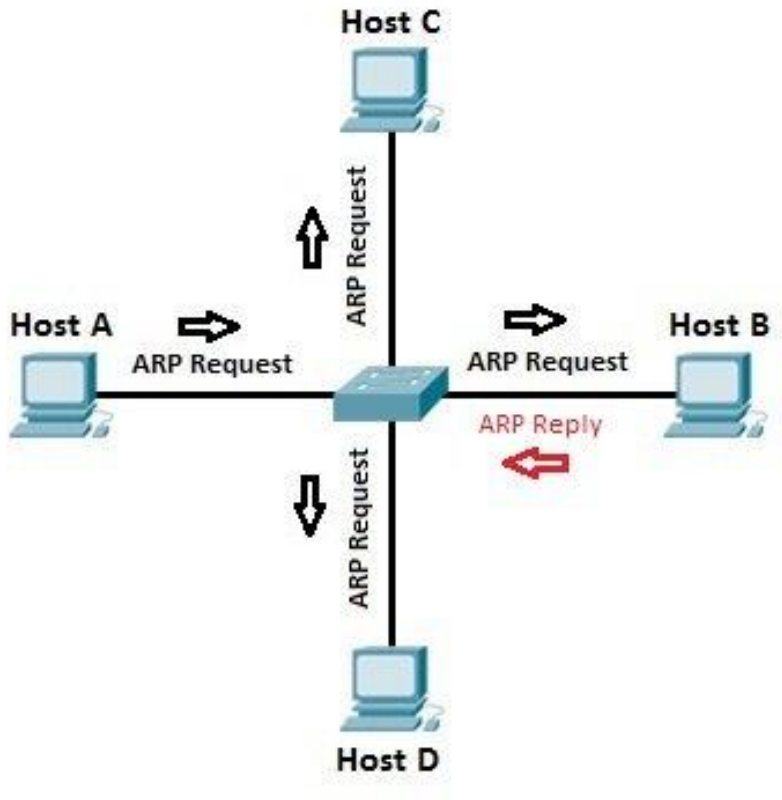
How ARP Works

- A host needs a destination's MAC address to build an Ethernet frame, but usually only knows its IP.
- ARP Request: broadcast — “Who has 192.168.1.5? Tell 192.168.1.10.”
- ARP Reply: unicast — the owner of that IP responds with its MAC address.
- The result is cached in the sender's ARP table to avoid repeating the broadcast for every packet.

ARP Cache — Example

IP Address	MAC Address	Type
192.168.1.1	AA:BB:CC:00:11:22	Dynamic
192.168.1.5	AA:BB:CC:00:33:44	Dynamic

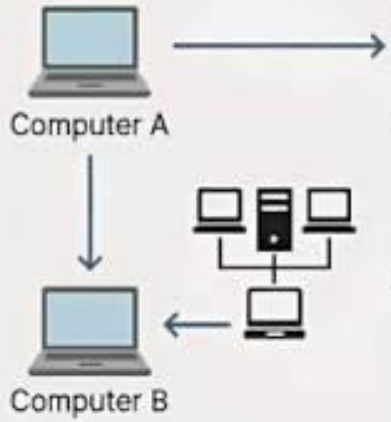
ARP Protocol



ARP

ARP (Address Resolution Protocol) is used to map an IP address to a MAC address within local network.

How ARP Works



1. ARP Request: Broadcasts „Who has IP address X.X.X.X?"
2. ARP Reply: Device with that IP address sends its MAC address
3. Caching: IP-MAC mapping is stored in the ARP cache

ARP Packet Types

- Request: Sent to obtain a MAC address for an IP address
- Reply: Sent in response to an ARP request
- Gratuitous ARP: Announces or updates a device's IP-to-MAC mapping

Security Risks and Mitigations

- ARP Spoofing: Attackers send fake ARP messages
- Use Dynamic ARP Inspection (DAI)
- Monitor network for suspicious ARP traffic

ARP: A Security Preview

- ARP has no authentication — a host will accept an unsolicited ARP reply and update its cache.
- ARP spoofing / poisoning: an attacker sends forged ARP replies to associate their own MAC with a victim's IP, enabling man-in-the-middle attacks.
- We'll cover detection and mitigation (e.g. dynamic ARP inspection) under Layer 2 attacks in Weeks 4–5.

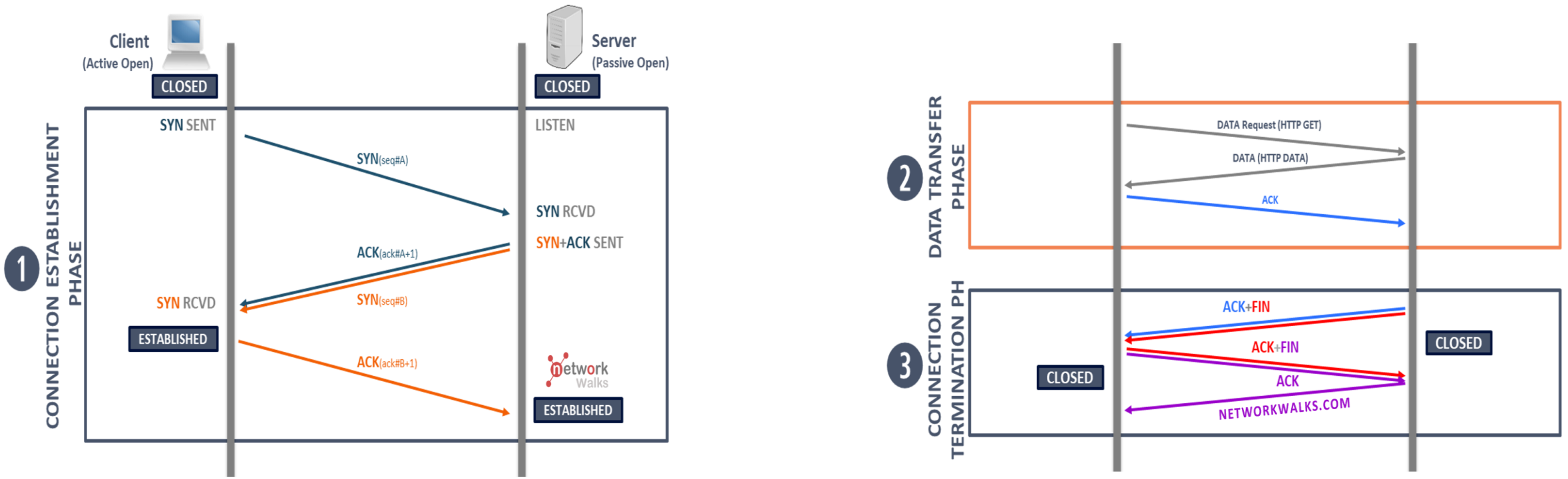
TCP

Reliable, connection-oriented transport

TCP Characteristics

- Connection-oriented: a session is established before data flows (three-way handshake).
- Reliable: lost segments are detected (via sequence numbers/ACKs) and retransmitted.
- Ordered: segments are reassembled in the correct order at the destination.
- Used where correctness matters more than speed: web (HTTP), email, file transfer.

TCP Communication



TCP Header — Key Fields

Field	Purpose
Source / Destination Port	Identify the sending and receiving application (socket)
Sequence / Acknowledgment Number	Track byte ordering and confirm receipt
Flags (SYN, ACK, FIN, RST)	Control connection setup, teardown, and reset
Window Size	Flow control — how much data the receiver can accept

Ports & Sockets

- A port (0–65535) identifies a specific application/service on a host.
- Well-known ports (0– 1023): HTTP=80, HTTPS=443, SSH=22, DNS=53.
- A socket = IP address + port — uniquely identifies one end of a connection.
- Firewalls commonly filter traffic by port — a preview of Week 7.

UDP

Fast, connectionless transport

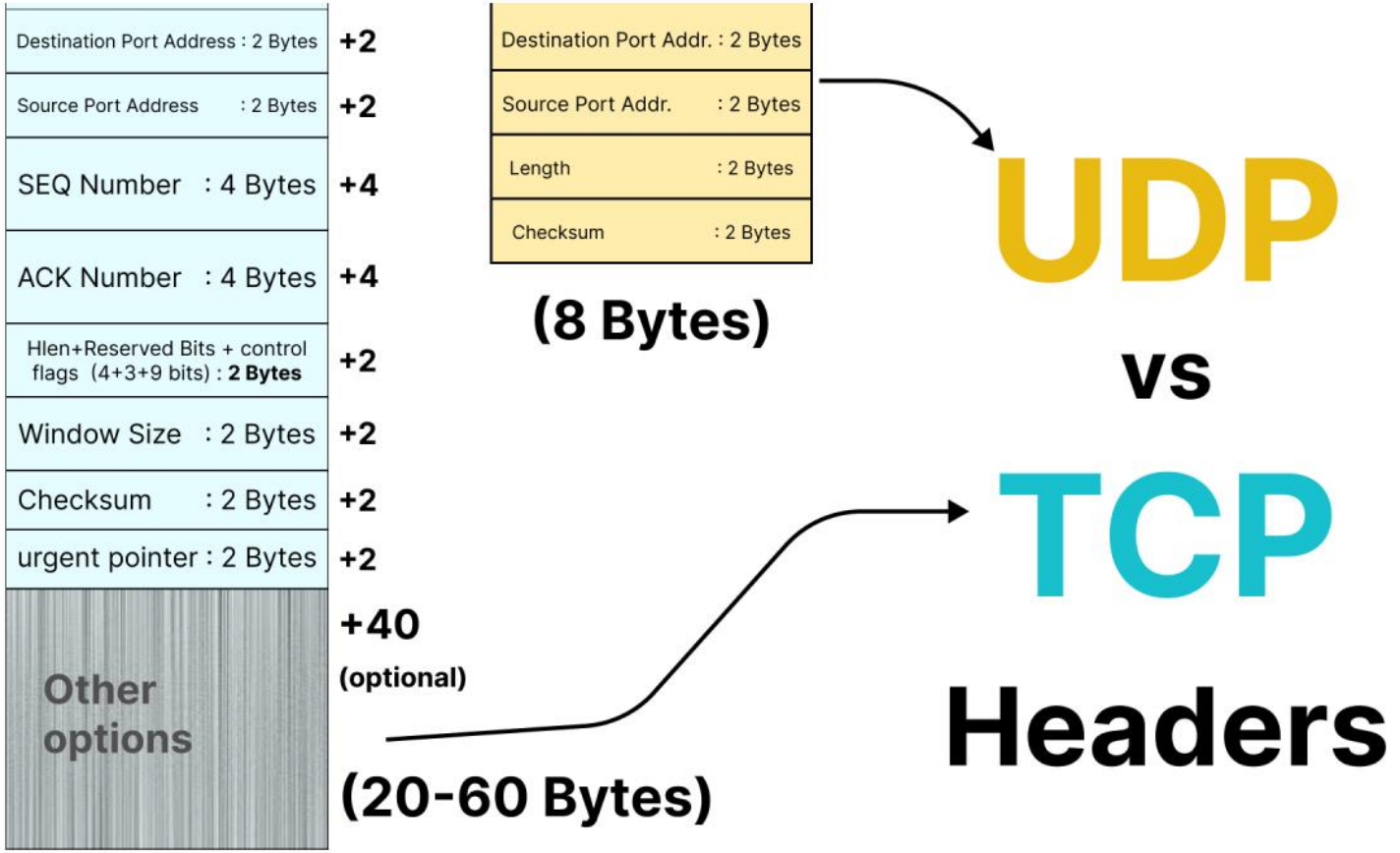
UDP Characteristics

- Connectionless: no handshake, no session state — just send.
- No reliability guarantees: no retransmission, no ordering, no flow control.
- Lower overhead than TCP — used where speed matters more than perfection: DNS, DHCP, VoIP, video streaming.
- That same lack of state makes UDP an easy vector for spoofing and amplification attacks.

UDP Header — Much Simpler

Field	Purpose
Source / Destination Port	Identify sending and receiving application
Length	Size of header + data
Checksum	Basic error detection (optional in IPv4)

UDP vs TCP Header



TCP vs UDP

	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Guaranteed delivery, ordered	Best-effort, no guarantees
Speed	Slower (handshake, ACKs)	Faster, minimal overhead
Typical use	Web, email, file transfer	DNS, DHCP, VoIP, streaming

TCP vs UDP

TCP	UDP
Secure	Unsecure
Connection-Oriented	Connectionless
Slow	Fast
Guaranteed Transmission	No Guarantee
Used by Critical Applications	Used by Real-Time Applications
Packet Reorder Mechanism	No Reorder Mechanism
Flow Control	No Flow Control
Advanced Error Checking	Basic Error Checking (Checksum)
20 Bytes Header	8 Bytes Header
Acknowledgement Mechanism	No Acknowledgement
Three-Way Handshake	No Handshake Mechanism
DNS, HTTPS, FTP, SMTP etc.	DNS, DHCP, TFTP, SNMP etc.

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Best Route To Your Dreams

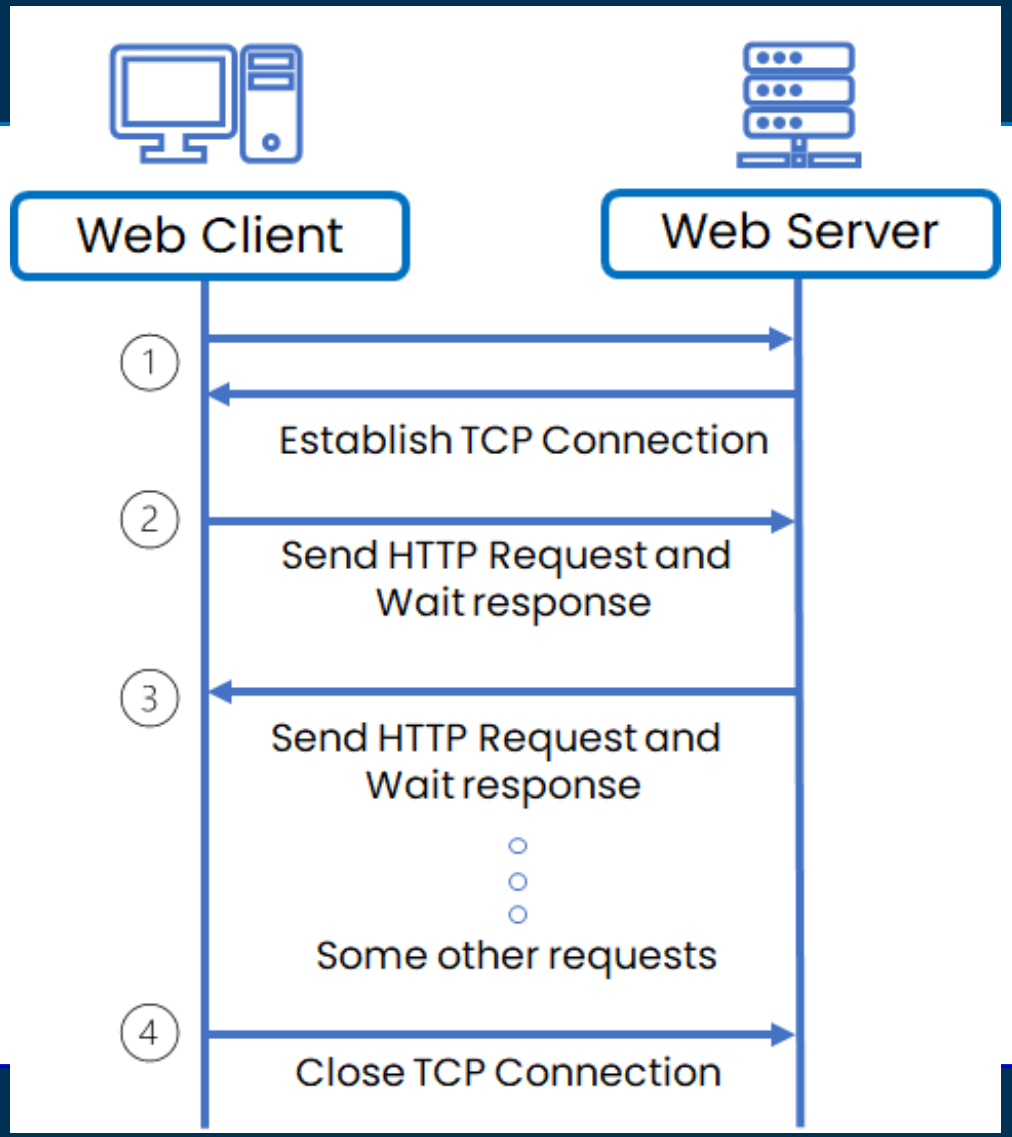
HTTP

The protocol behind the web

HTTP Basics

- Request/response protocol: a client sends a request, the server sends a response.
- Common methods: GET (retrieve), POST (submit data), PUT (update), DELETE (remove).
- Runs over TCP, traditionally on port 80.
- Stateless by design — each request is independent (cookies/sessions add state on top).

HTTP Protocol



HTTP vs HTTPS

	HTTP	HTTPS
Port	80	443
Encryption	None — plaintext	TLS/SSL encrypted
Data exposed to eavesdroppers?	Yes	No (in transit)

HTTP: A Security Preview

- Plaintext HTTP exposes credentials, cookies, and content to anyone who can see the traffic — relevant to Layer 3/4 attacks like sniffing.
- HTTPS (TLS) provides confidentiality and integrity, but the certificate trust model itself can be attacked (covered under Digital Certificates, Week 12).
- Common headers and misconfigurations here become relevant again when we reach cryptography and PKI.

Summary

- Layered architecture (OSI/TCP-IP) gives us a shared vocabulary for where an attack or defense operates.
- Switching and routing move frames/packets — and every hop is a place security policy can be enforced.
- Ethernet, IP, DHCP, and ARP move data at Layers 1–3 — mostly without built-in authentication, which is why they're common attack targets.
- TCP and UDP shape how attacks like SYN floods and spoofing work at Layer 4.
- HTTP/HTTPS shows why encryption matters at the application layer.

This Week's Lab

- Part A — Incident case studies: analyse real, recent network security breaches and answer scenario questions about vulnerability, impact, and mitigation.
- Part B — Job market scan: review local and international entry-level cybersecurity job ads and tally how many require networking/network security skills.
- See the Week 1 Lab Guide for full instructions.

References

- Kaufman, Perlman, Speciner & Perlner, Network Security: Private Communication in a Public World, 3rd Ed., Prentice Hall.
- Santos & Stuppi, CCNA Security 210-260 Official Cert Guide, 1st Ed., Cisco Press.
- Cisco Networking Academy — Introduction to Networks curriculum.